

SUSPENSION AND STEERING ANALYSIS REPORT

ABSTRACT

Suspension and steering system are the integral parts of any vehicle. The main components of suspension and steering are wheel knuckle, control arms (A-arms), rocker, wheel hub and rack and pinion. The suspension maintains stability while the steering ensures directional control. Faulty components in either system can lead to poor handling, uneven tire wear, or even loss of control. Hence it is necessary to analyze these components, since they directly affect the stability of the vehicle and comfort and safety of the driver.

The following report presents the FEA analysis of the above-mentioned components. The forces and moments acting on the components were simulated and the resulting Von-Mises stress, total deformation and factor of safety were obtained. Von-mises stresses were chosen for the analysis since all the components being analyzed are made of ductile materials and Von-mises failure theory is the most suitable for ductile materials. The factor of safety was calculated to ensure that the induced stresses are well below the safety limits. The FEA analysis was performed using Ansys and the obtained results were analyzed to assess the safety of the components.

WHEEL KNUCKLE ANALYSIS

Symbols and their values

Mass of vehicle, M	250 kg
Mass of front of vehicle, M _f	114 kg
Mass of rear of vehicle, M _r	136 kg
Wheelbase, L	1625 mm
Front trackwidth, T _f	1200 mm
Rear trackwidth, T _r	1150 mm
Height of center of mass above ground, h	325 mm
Speed while cornering, v	45 km/h = 12.5 m/s
Coefficient of friction, μ	0.72
Radius of tyre, r	228.6 mm

Mass of wheel, m	5 kg
Cornering radius, R	3 m
Distance of hub centre to lower A arm point in front wheel assembly, a	110.9 mm
Distance of hub centre to lower A arm point in front wheel assembly, b	110.9 mm
Scrub radius, r _s	63.77 mm
Length of steering arm, l	45 mm
Distance b/w centre of brake mounting and centre of rear knuckle, c	55 mm

Objective

The knuckle is a fundamental component of the suspension system, acting as the connection point between the suspension arms and the wheel assembly. It needs to connect the car's suspension, brakes, and wheel hub while handling the huge forces from racing.

The knuckle must be strong and stiff enough to withstand a variety of loads, including vehicle weight, braking forces and cornering forces. Besides this, the knuckle should also be light in weight. Making the knuckle lightweight helps the suspension react faster and improves grip. To achieve these objectives, strong and lightweight materials like aluminum alloys are chosen.

Design specification

Aluminum 6061 T6 alloy was used for fabricating the knuckles. This material was chosen for its excellent strength-to-weight ratio and good corrosion resistance while allowing a lightweight design.

The height of the front knuckle is 221.79 mm, the length is 94.66 mm, and the width is 31.52 mm.

The height of the rear knuckle is 220.23 mm, the length is 104.45 mm, and the width is 39.88 mm.

The properties of Aluminum 6061 T6 alloy are:

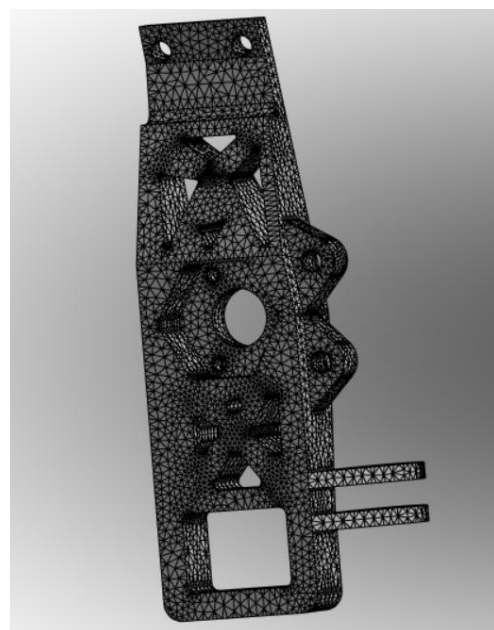
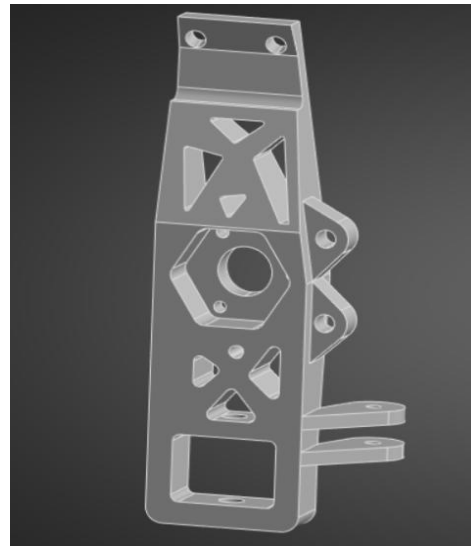
Young's Modulus	200 GPa
Yield Tensile Strength	250 MPa
Ultimate Tensile Strength	460 MPa
Poisson's ratio	0.3

FRONT KNUCKLE ANALYSIS

Analysis Methodology

Finite Element Analysis (FEA) was performed on the knuckle using Ansys software to simulate the effect of the forces and moments which are acting on the front knuckle.

Meshing was done with adaptive sizing and element size was set to 1 mm. Tetrahedral elements were chosen for meshing since a tetrahedral mesh has the advantage of being able to well approximate the surface contour. The number of elements in the mesh were 1,35,329 and the number of nodes were 36,624.



Forces Calculation

Braking forces

Deceleration (acc) = $1.2g = 11.77 \text{ m/s}^2$

Weight on one wheel, $F_w = M_f \cdot g / 2 = 559.2 \text{ N}$

Longitudinal load transfer on one wheel during braking = $M \cdot \text{acc} \cdot h / (2 \cdot L) = 294.25 \text{ N}$

Total vertical load (F_v) = $559.2 + 294.25 = 853.45 \text{ N}$

Frictional force (f) = $\mu \cdot F_v = 614.48 \text{ N}$

Braking torque (T_b) = $f \cdot r = 140.47 \text{ N-m}$

$F_1 = (f \cdot a - T_b) / (a + b) = -326.09 \text{ N}$

$F_2 = f - F_1 = 940.59 \text{ N}$

Cornering forces

Vertical force due to centrifugal couple (F_c) = $(M \cdot v^2 \cdot h) / (2 \cdot R \cdot T_f) = 1763.2 \text{ N}$

Vertical force due to gyroscopic effect (F_g) = $4 \cdot (m \cdot v^2 \cdot r^2) / (r \cdot 2 \cdot R) = 119.06 \text{ N}$

Net vertical load (F_{net}) = $F_w + F_c + F_g = 2441.5 \text{ N}$

Lateral force (F_L) = $\mu \cdot F_{\text{net}} = 1757.9 \text{ N}$

$F_3 = F_L \cdot (r - a) / (a + b) = 932.82 \text{ N}$

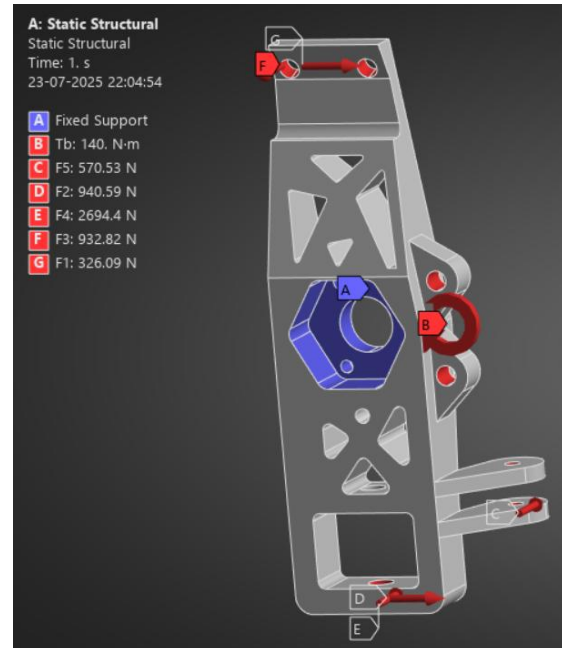
$F_4 = F_L + F_2 = 2698.4 \text{ N}$

Forces on Steering arm

Frictional force (f_o) = $\mu \cdot F_w = 402.6 \text{ N}$

Torque about steering axis (T_s) = $f_o \cdot r_s = 25.67 \text{ N-m}$

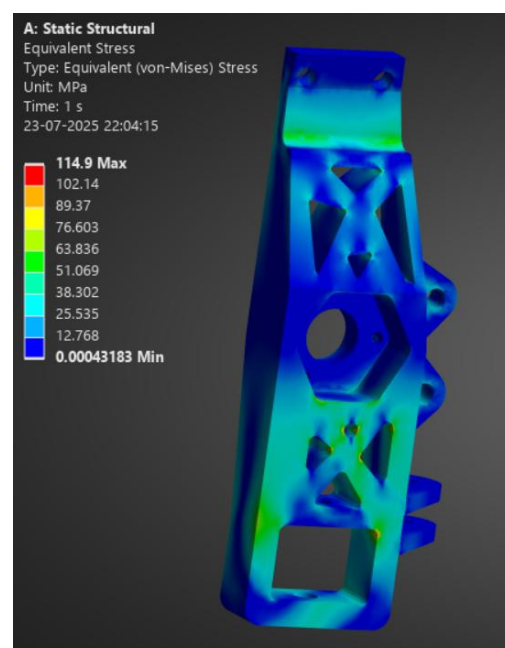
Force on tie rod (F_5) = $T_s / l = 570.53 \text{ N}$



Results

The maximum equivalent von mises stress was 114.9 MPa . The maximum deformation was 0.228 mm . The factor of safety exceeds 2.26 .

This indicates that the front knuckle design is safe and capable of handling the forces encountered during operation without risk of failure.

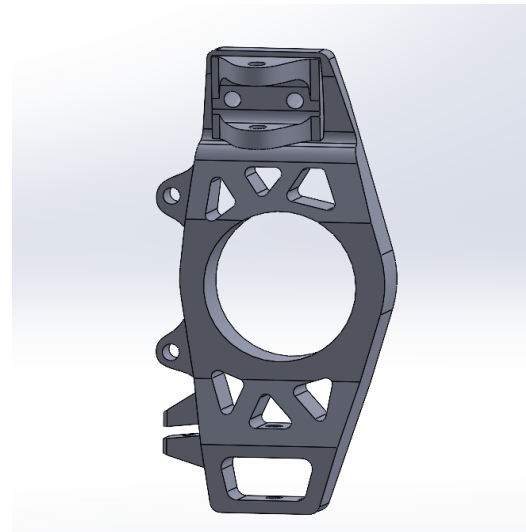
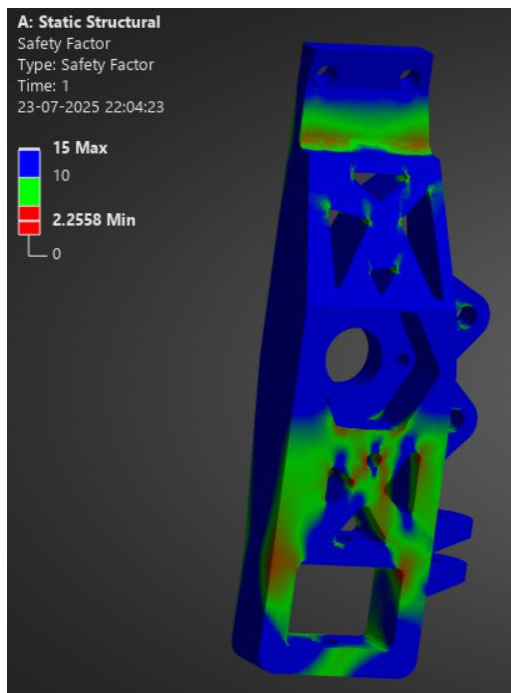
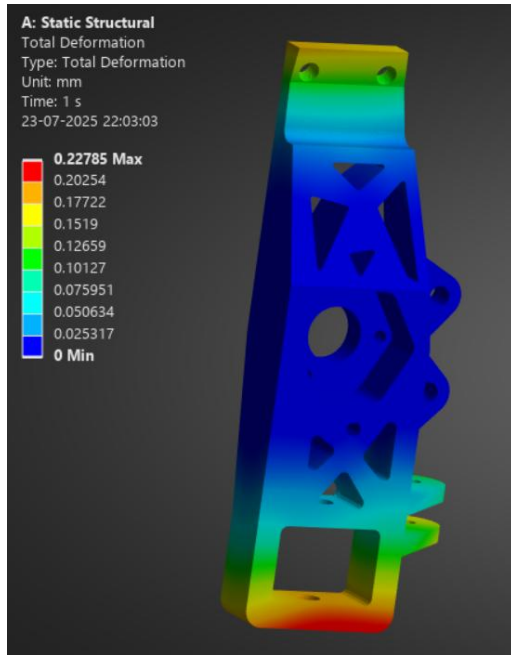


REAR KNUCKLE ANALYSIS

Analysis Methodology

Finite Element Analysis (FEA) was performed on the knuckle using Ansys software to simulate the effect of the forces which are acting on the rear knuckle.

Meshing was done with adaptive sizing and element size was set to 1 mm. Tetrahedral elements were chosen for meshing since a tetrahedral mesh has the advantage of being able to well approximate the surface contour. The number of elements in the mesh were 1,58,770 and the number of nodes were 2,74,399.



Forces Calculation

Acceleration (acc) = $1.2g = 11.77 \text{ m/s}^2$

Weight on one rear knuckle (W_r) = $M_r \cdot g / 2 = 667.1 \text{ N}$

Longitudinal load transfer during accelerating = $M \cdot \text{acc} \cdot h / (2 \cdot L) = 294.25 \text{ N}$

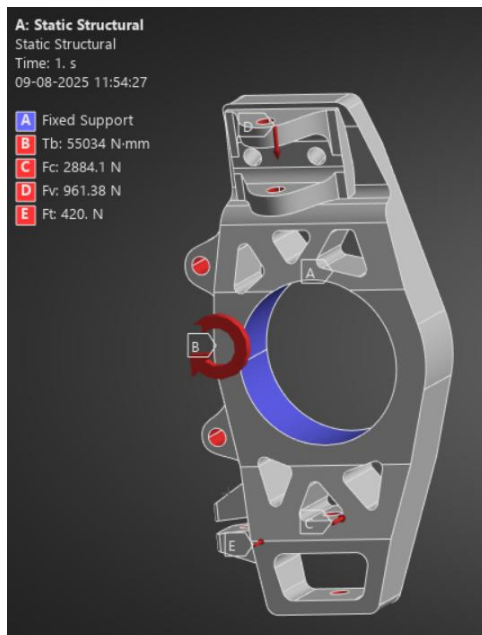
Total vertical load (F_v) = $667.1 + 294.25 = 961.38 \text{ N}$

Cornering force (F_c) = $3 \cdot F_v = 2884.1 \text{ N}$

Braking force (F_b) = $1.5 \cdot W_r = 1000.6 \text{ N}$

Braking moment (T_b) = $F_b \cdot c = 55.03 \text{ N-m}$

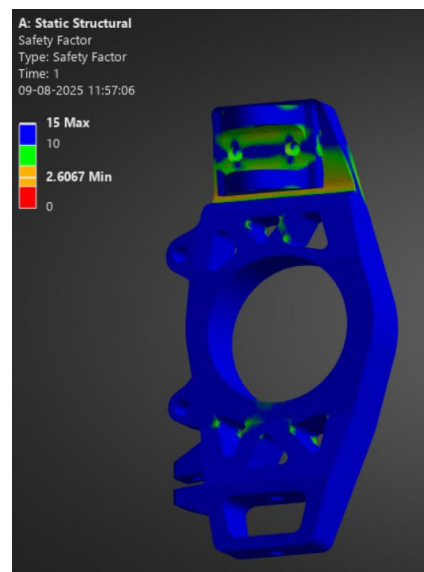
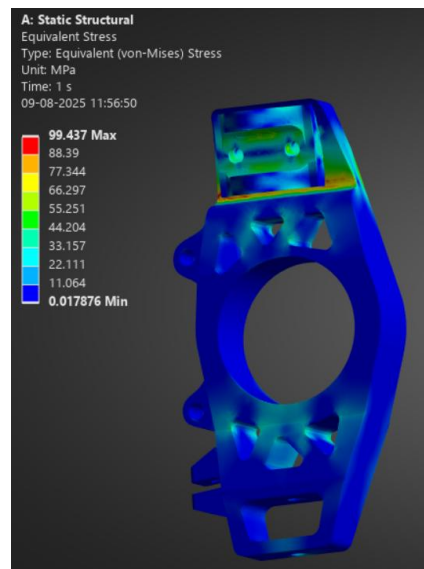
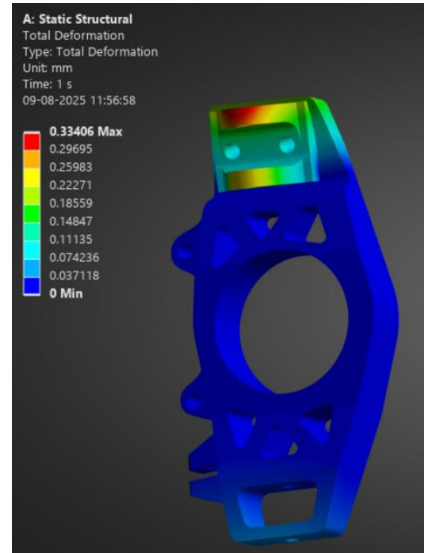
Force on tie-rod (F_t) = $\mu \cdot F_v = 420 \text{ N}$



Results

The maximum equivalent von mises stress was 99.437 MPa . The maximum deformation was 0.334 mm . The factor of safety exceeds 2.61 .

This indicates that the rear knuckle design is safe and capable of handling the forces encountered during operation without risk of failure.



A-ARMS ANALYSIS

Objective

The double wishbone suspension system is a suspension design which is widely used in high-performance race cars. It provides a high degree of control over the wheel's movement and helps maintain the wheel parameters like camber for optimum performance. The suspension has one upper wishbone and one lower wishbone. Each wishbone has two arms.

The weight of the suspension components, wheel, and tire should be as low as possible. A lighter unsprung mass allows the suspension to react more quickly to road imperfections, improving grip and handling. The wishbone material must be strong and stiff and must be able to resist deformation while handling forces like weight, longitudinal load transfer, lateral load transfer.

Design specification

AISI 4130 chromoly steel was used for fabricating the A-arms. A-arms made from this material can endure high stresses while remaining lightweight.

The properties of AISI 4130 Chromoly steel are:

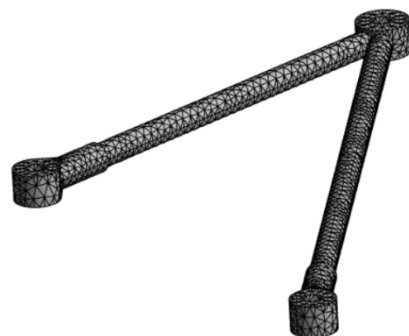
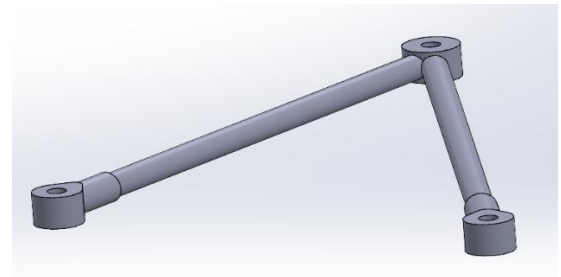
Young's Modulus	210 GPa
Yield Tensile Strength	460 MPa
Ultimate Tensile Strength	560 MPa
Poisson's ratio	0.27

FRONT UPPER A-ARM ANALYSIS

Analysis Methodology

Finite Element Analysis (FEA) was performed on the A-arm using Ansys software to simulate the effect of the forces which are acting on it.

Meshing was done with adaptive sizing and element size was set to 1 mm. Tetrahedral elements were chosen for meshing since a tetrahedral mesh has the advantage of being able to well approximate the surface contour. Also, the meshing time for curved geometries is less when using tetrahedral mesh. The number of elements in the mesh were 71,455 and the number of nodes were 1,24,303.



Forces Calculation

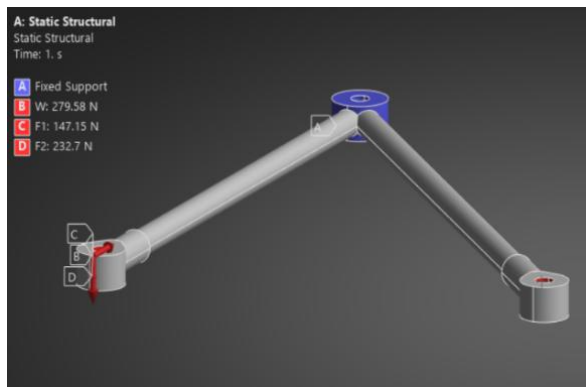
Deceleration during braking ($acc1$) = $1.2g$ = 11.77 m/s^2

Acceleration during cornering ($acc2$) = $1.4g$ = 13.73 m/s^2

Weight on one front upper wishbone (Wf) = $(Mf \cdot g)/4 = 279.58 \text{ N}$

Longitudinal load transfer during accelerating ($F1$) = $(M \cdot acc1 \cdot h)/(4 \cdot L) = 147.15 \text{ N}$

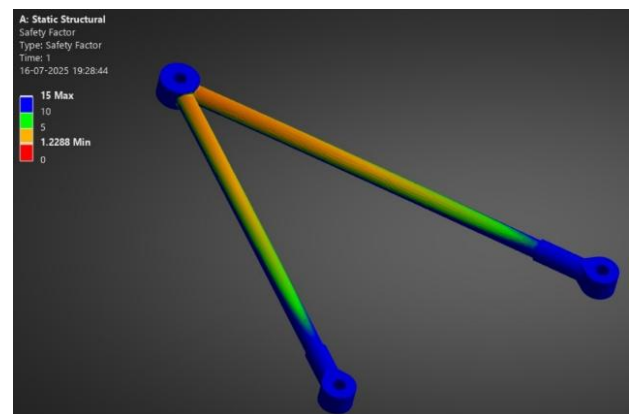
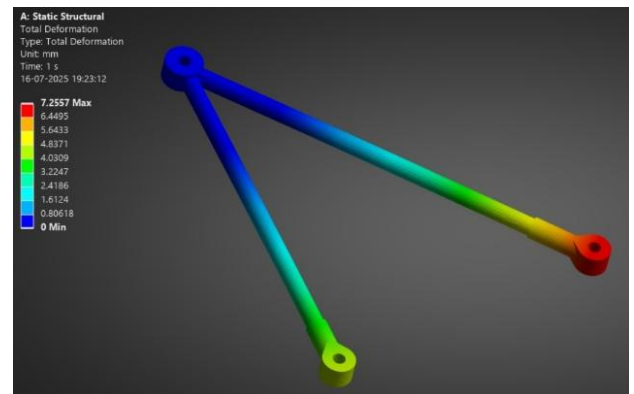
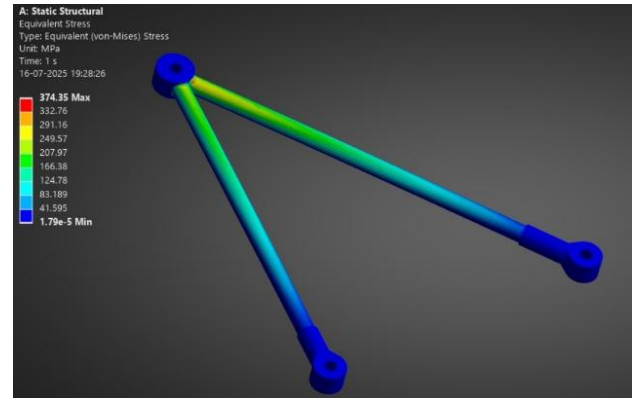
Lateral load transfer during cornering ($F2$) = $(M \cdot acc2 \cdot h)/(4 \cdot t) = 232.7 \text{ N}$



Results

The maximum equivalent von mises stress was 374.35 MPa. The maximum deformation was 7.26 mm. The factor of safety exceeds 1.23.

This indicates that the front upper wishbone design is safe and capable of handling the forces encountered during operation without risk of failure.



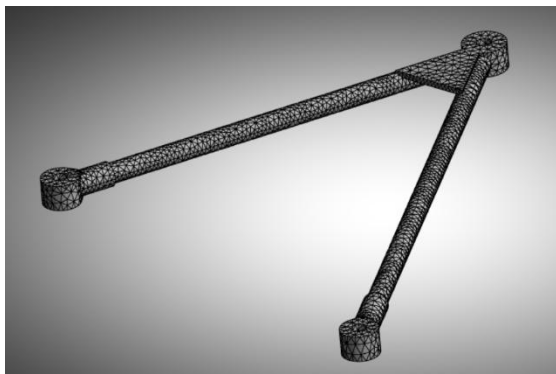
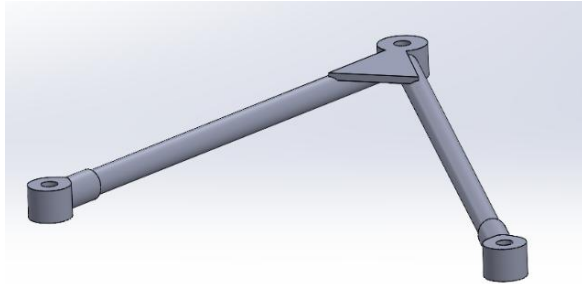
FRONT LOWER A-ARM ANALYSIS

Analysis Methodology

Finite Element Analysis (FEA) was performed on the A-arm using Ansys software to simulate the effect of the forces which are acting on it.

Meshing was done with adaptive sizing and element size was set to 1 mm. Tetrahedral

elements were chosen for meshing since a tetrahedral mesh has the advantage of being able to well approximate the surface contour. Also, the meshing time for curved geometries is less when using tetrahedral mesh. The number of elements in the mesh were 72,814 and the number of nodes were 1,26,979.



Forces Calculation

Deceleration during braking ($acc1$) = $1.2g$ = 11.77 m/s^2

Acceleration during cornering ($acc2$) = $1.4g$ = 13.73 m/s^2

Weight on one front lower wishbone (W_f) = $M_f \cdot g / 4 = 279.58 \text{ N}$

Longitudinal load transfer during accelerating ($F1$) = $M \cdot acc1 \cdot h / (4 \cdot L) = 147.15 \text{ N}$

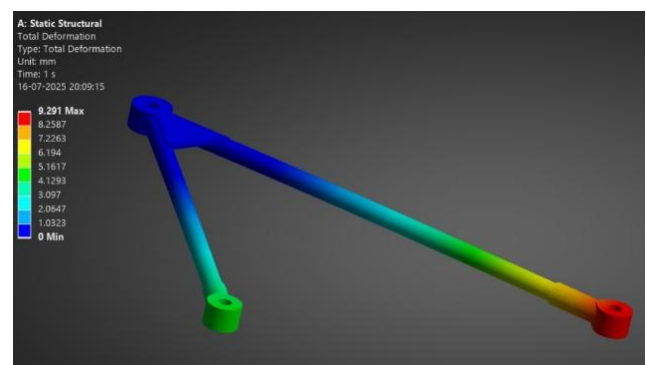
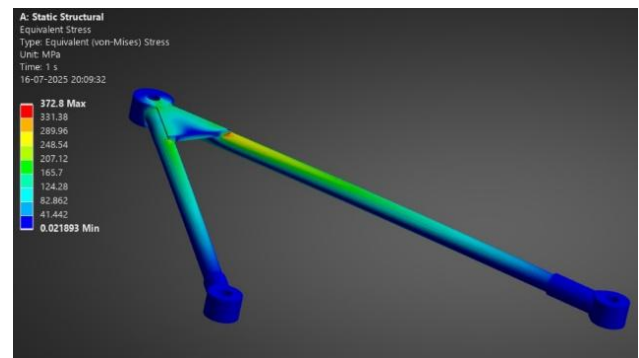
Lateral load transfer during cornering ($F2$) = $M \cdot acc2 \cdot h / (4 \cdot t) = 232.7 \text{ N}$

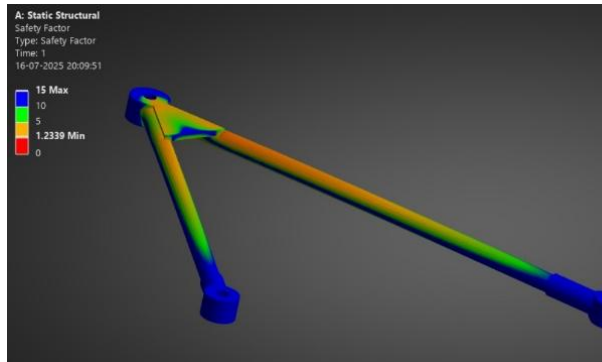


Results

The maximum equivalent von mises stress was 372.8 MPa. The maximum deformation was 9.29 mm. The factor of safety exceeds 1.23.

This indicates that the front lower wishbone design is safe and capable of handling the forces encountered during operation without risk of failure.





REAR LOWER A-ARM ANALYSIS

Analysis Methodology

Finite Element Analysis (FEA) was performed on the A-arm using Ansys software to simulate the effect of the forces which are acting on it.

Meshing was done with adaptive sizing and element size was set to 1 mm. Tetrahedral elements were chosen for meshing since a tetrahedral mesh has the advantage of being able to well approximate the surface contour. Also, the meshing time for curved geometries is less when using tetrahedral mesh. The number of elements in the mesh were 64,892 and the number of nodes were 1,13,367.



Forces Calculation

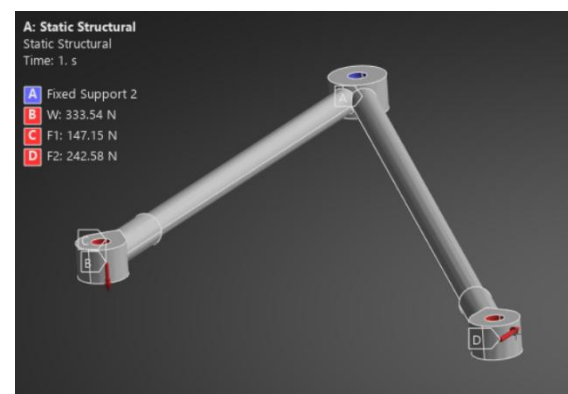
Acceleration ($acc1$) = $1.2g = 11.77 \text{ m/s}^2$

Acceleration during cornering ($acc2$) = $1.4g = 13.73 \text{ m/s}^2$

Weight on one rear lower wishbone (W_r) = $M_r * g / 4 = 333.54 \text{ N}$

Longitudinal load transfer during accelerating ($F1$) = $M * acc1 * h / (4 * L) = 147.15 \text{ N}$

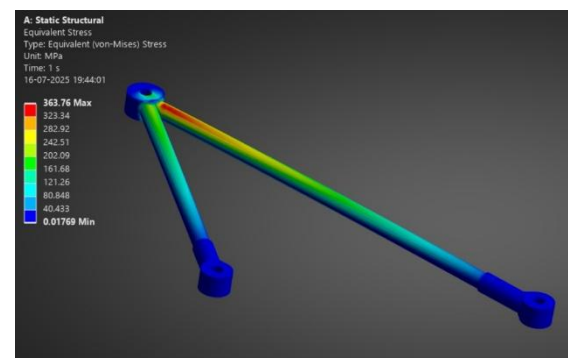
Lateral load transfer during cornering ($F2$) = $M * acc2 * h / (4 * t) = 242.58 \text{ N}$

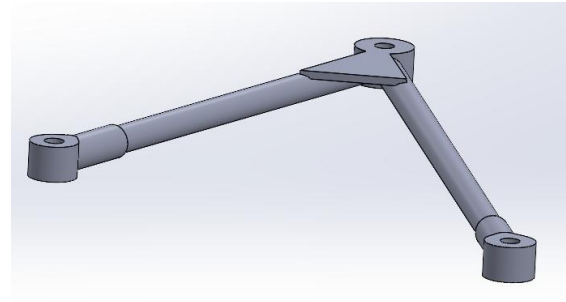
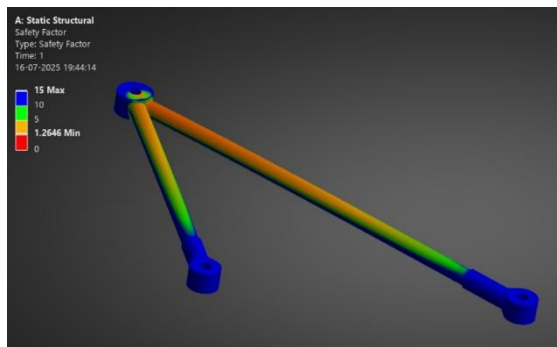
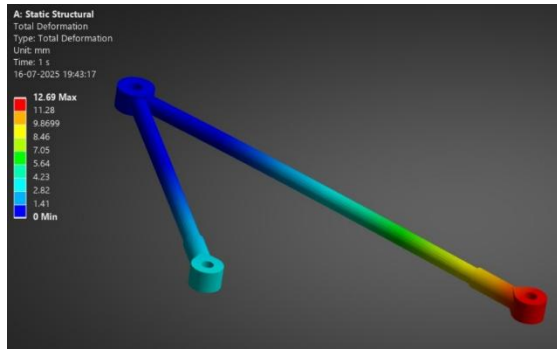


Results

The maximum equivalent von mises stress was 363.76 MPa. The maximum deformation was 12.69 mm. The factor of safety exceeds 1.26.

This indicates that the rear lower wishbone design is safe and capable of handling the forces encountered during operation without risk of failure.





REAR UPPER A-ARM ANALYSIS

Analysis Methodology

Finite Element Analysis (FEA) was performed on the A-arm using Ansys software to simulate the effect of the forces which are acting on it.

Meshing was done with adaptive sizing and element size was set to 1 mm. Tetrahedral elements were chosen for meshing since a tetrahedral mesh has the advantage of being able to well approximate the surface contour. Also, the meshing time for curved geometries is less when using tetrahedral mesh. The number of elements in the mesh were 67,344 and the number of nodes were 1,17,281.

Forces Calculation

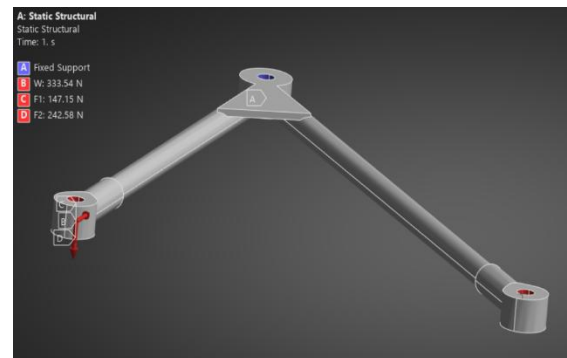
Acceleration ($acc1$) = $1.2g = 11.77 \text{ m/s}^2$

Acceleration during cornering ($acc2$) = $1.4g = 13.73 \text{ m/s}^2$

Weight on one rear upper wishbone (W_r) = $M_r \cdot g / 4 = 333.54 \text{ N}$

Longitudinal load transfer during accelerating ($F1$) = $M \cdot acc1 \cdot h / (4 \cdot L) = 147.15 \text{ N}$

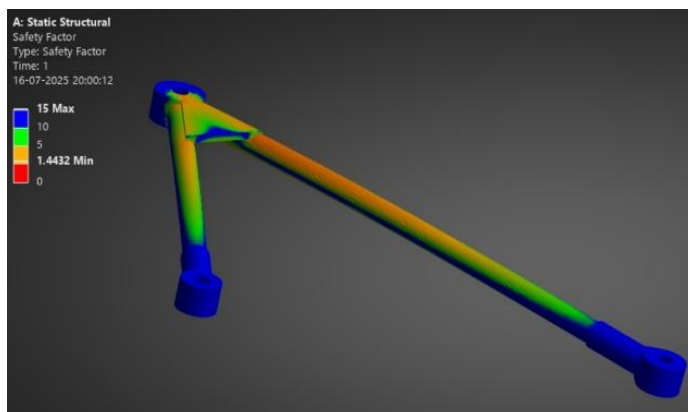
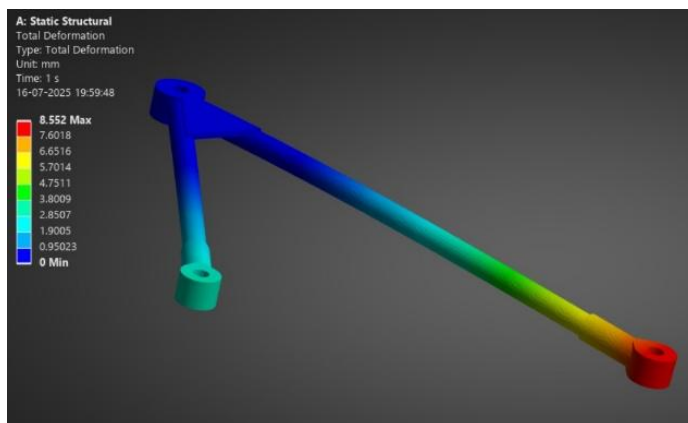
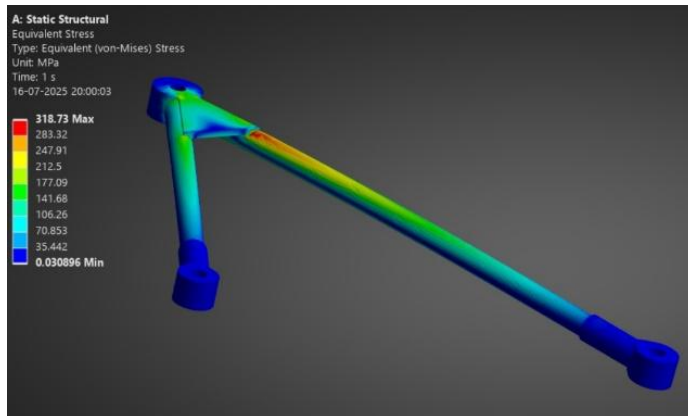
Lateral load transfer during cornering ($F2$) = $M \cdot acc2 \cdot h / (4 \cdot t) = 242.58 \text{ N}$



Results

The maximum equivalent von mises stress was 318.73 MPa. The maximum deformation was 8.55 mm. The factor of safety exceeds 1.44.

This indicates that the rear upper wishbone design is safe and capable of handling the forces encountered during operation without risk of failure.



ROCKER ANALYSIS

Introduction

A rocker is a key component in pushrod suspension systems, commonly used in high-performance vehicles like Formula cars. It acts as a mechanical link that transfers vertical forces from the pushrod, connected to the wheel hub, to the spring-damper unit. The rocker rotates about a pivot point, allowing the suspension to manage road irregularities while controlling wheel movement. It helps in packaging flexibility, provides motion ratio tuning, and reduces unsprung mass, making the suspension system more responsive and adjustable to various driving conditions.

Material Selection

The rocker arms are made from mild steel because it offers good strength, toughness, and excellent machinability, making it suitable for manufacturing and welding.

Density	7850 kg/m ³
Yield strength	250 MPA
Tensile strength	475 MPA
Poisson ratio	0.3
Young Modulus	200 GPa

Forces Calculation

In a rocker suspension system, the bump force is the vertical load acting on the wheel hub during bumps or cornering, calculated by dividing the dynamic load by the number of wheels. This bump force is transmitted through the pushrod, where it converts into an axial force depending on the pushrod angle. The pushrod force is determined by dividing the bump force by the sine of the pushrod angle.

The rocker arm pivots and balances forces between the pushrod and the damper. By applying moment equilibrium around the rocker pivot, the product of pushrod force and its distance from the pivot equals the product of spring/damper force and its distance from the pivot. This ensures proper load transfer and suspension efficiency.

For rear rocker

a) Bump Force

$$\text{Wheel rate} = \text{Spring rate} \times (\text{motion ratio})^2$$

$$\text{Spring rate for rear} = 14.8 \text{ N/mm}$$

$$\text{Motion ratio} = 1.4$$

$$\text{Wheel rate} = 14.8 \times (1.4)^2 = 29.008 \text{ N/mm}$$

$$\begin{aligned}\text{Rear Bump force} &= \text{wheel rate} \times \text{bump travel} \\ &= 29.008 \times 25.4 \\ &= 736.80 \text{ N/mm}\end{aligned}$$

b) Force on pushrod rocker

$$\begin{aligned}\text{Force pushrod rocker} &= \text{bump force} \times \cos(45^\circ) \\ &= 736.80 \times \cos(45^\circ) \\ &= 603.55 \text{ N}\end{aligned}$$

c) Force on rocker Damper end

$$\begin{aligned}\text{Rocker motion ratio} &= (L_1/L_2) \times \sin(\theta) \\ &= (54.11/86.05) \times \sin(80^\circ) \\ &= 0.61\end{aligned}$$

$$\begin{aligned}\text{Force on rocker end} &= \text{force on pushrod} \times \text{RMR} \\ &= 603.55 \times 0.61 \\ &= 368.166 \text{ N}\end{aligned}$$

For Front rocker

a) Bump Force

$$\text{Wheel rate} = \text{Spring rate} \times (\text{motion ratio})^2$$

$$\text{Spring rate for front} = 11.8 \text{ N/mm}$$

$$\text{Motion ratio} = 1.4$$

$$\text{Wheel rate} = 11.8 \times (1.4)^2$$

$$= 23.128 \text{ N/mm}$$

$$\text{Rear Bump force} = \text{wheel rate} \times \text{Bump travel}$$

$$= 23.128 \times 25.4$$

$$= 588.774 \text{ N/mm}$$

b) Force on pushrod rocker

$$\text{Force on pushrod rocker} = \text{bump force} \times \cos(5^\circ)$$

$$= 588.774 \times \cos(5^\circ)$$

$$= 586.53 \text{ N}$$

c) Force on rocker Damper end

$$\text{Rocker motion ratio} = (L1/L2) \times \sin(\theta)$$

$$= (137/145) \times \sin(19^\circ)$$

$$= 0.30$$

$$\text{Force on rocker end} = \text{force on pushrod} \times \text{RMR}$$

$$= 586.53 \times 0.30$$

$$= 179.6612 \text{ N}$$

Finite element analysis

Finite Element Analysis (FEA) of a rocker in a suspension system is performed to evaluate its structural strength, stress distribution, and deformation behaviour under applied loads. The primary loads considered are the pushrod force, which comes from bump loads at a specific angle, and the spring/damper reaction force from the opposite side of the rocker.

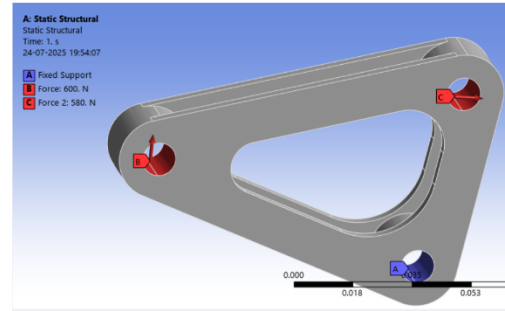


Fig 1.1 Structural forces on rear rocker

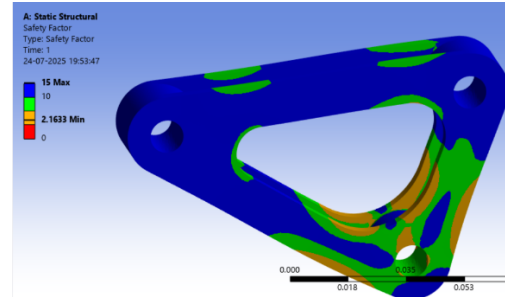


Fig 1.2 Factor of safety rear rocker

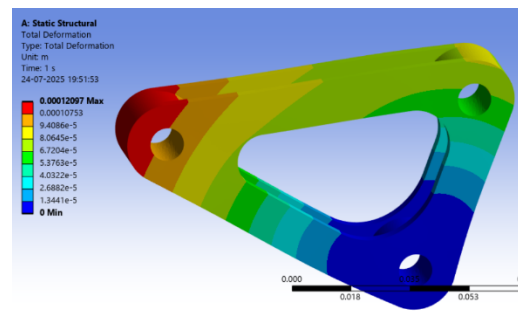


Fig 1.3 Total deformation on rear rocker

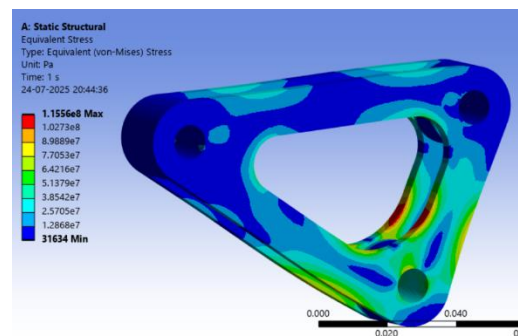


Fig 1.4 Equivalent stress on rear rocker

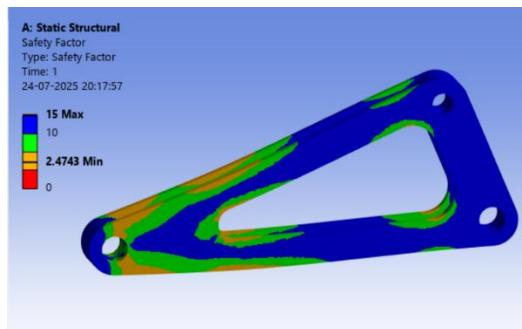


Fig 1.5 Factor of safety on front rocker

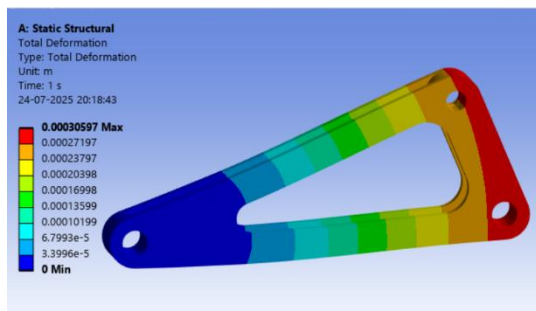


Fig 1.6 Total deformation on front rocker

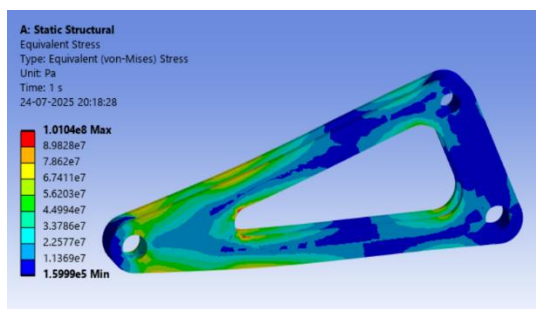


Fig 1.7 Equivalent stress on front rocker

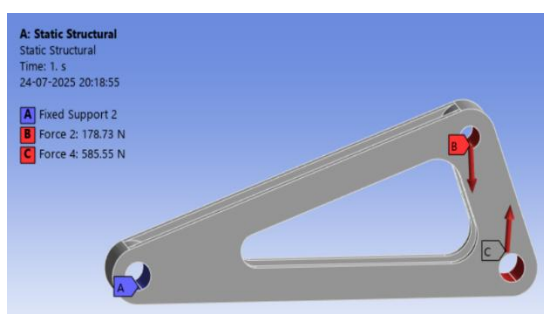


Fig 1.8 Structural forces on front rocker

Result

Rear Rocker:

- Total Deformation: *0.120 mm*
- Maximum Equivalent (Von Mises) Stress: *115.5 MPa*
- Factor of Safety (FOS): *2.163*

Front Rocker:

- Total Deformation: *0.305mm*
- Maximum Equivalent (Von Mises) Stress: *101.04 MPa*
- Factor of Safety (FOS): *2.47*

Significance of Results

- Total Deformation for both front and rear rockers is under 0.4mm, indicating excellent rigidity and minimal flex under load.
- The maximum stress values (Rear: 155.5 MPa, Front: 101.04 MPa) are well below the yield strength of mild steel (250 MPa), which means no plastic deformation will occur during operation.
- The Factor of Safety is above 2 for both rockers, ensuring the design can handle loads beyond expected limits with sufficient margin.

Conclusion

From the finite element analysis of the rocker, it was observed that the maximum induced stress remains below the yield strength of the material (mild steel). The deformation levels are within safe limits, ensuring no excessive flexing under maximum load conditions. Additionally, the Factor of Safety (FOS) is satisfactory, indicating the rocker design is structurally safe and capable of withstanding the applied bump and pushrod forces without risk of failure.

WHEEL HUB ANALYSIS

Introduction

In high-performance racing such as Formula SAE, the goal is to achieve a superior power-to-weight ratio without compromising on safety or reliability. Hub plays a crucial role in this balance. Hub is responsible for:

- Transferring loads between the suspension and wheel assembly,
- Withstanding vertical loads from the vehicle's weight,
- Enduring lateral forces during cornering,
- Handling torque from braking and drivetrain input.

Methodology

Each component was modelled using SolidWorks 2023. During the CAD process, particular attention was given to ensuring structural symmetry, ease of assembly, and compatibility with existing drivetrain, suspension, and braking subsystems.

The rear hub was designed with a central spline bore for the driveshaft and flanges for mounting the brake disc and wheel. The front hub incorporates precise interfaces for bearings, rotor mounting points, and wheel studs.

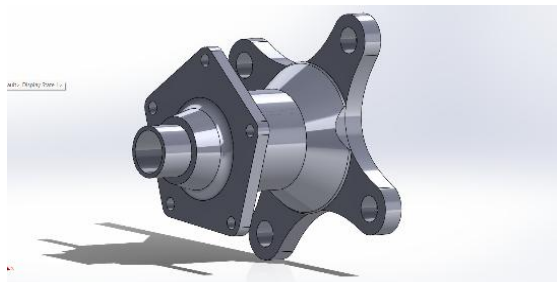


Fig 25.2 Rear Hub

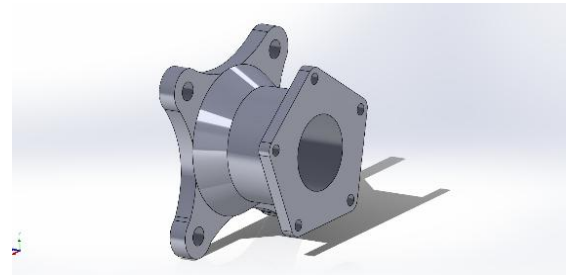


Fig 25.3 Front hub

Material Selection

Material choice was crucial to achieving a high strength-to-weight ratio. For the front and rear hubs, Aluminium 7075-T6 was selected due to its excellent mechanical properties and machinability.

Density	2810 kg/m ³
Yield Strength	500 MPa
Ultimate tensile strength	570 MPa
Young Modulus	71 GPa
Poisson ratio	0.33

Calculation Analysis

The strength and durability of race car components depend significantly on the accurate estimation of dynamic forces during various operating conditions. To ensure the structural reliability of the rear hub and front hub calculations were performed to determine the magnitude of loads acting under braking, cornering, suspension compression (bump).

The total forces applied in CAE simulations were derived from the following major load contributors:

1. Braking torque

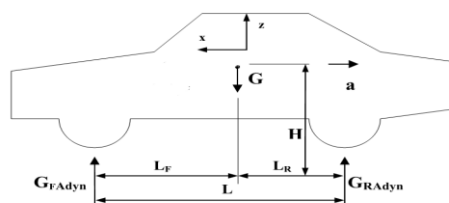
2. Cornering (lateral) forces
3. Bump (suspension compression) forces

Vehicle Specification

Parameter	Value
Wheelbase(L)	1626 mm
Trackwidth	1200 mm (front) 1150 mm (rear)
Cornering radius	3m
Scrub radius	63.77 mm
Height of centre of gravity(H)	325 mm
Turning radius	2.5 m
Tyre radius (Hoosier)	0.311 m
Length of steering arm	45 mm
Mass of vehicle with driver(M)	250 kg

Braking Torque

When braking occurs, longitudinal weight transfer increases the vertical load on the front wheels, generating a reaction force at the calliper mounting points. This leads to a torque around the spindle centre.



25.5 FBD of vehicle

- Vehicle velocity (u): 11.11 m/s
- Braking distance (S): 4.1 m

- Using the equation $v^2 = u^2 + 2 \times a \times S$, the deceleration is calculated as:

$$0 = (12.5)^2 + 2 \times a \times (4.57) \text{ where } a = -15.05 \text{ m/s}^2$$

Sprung + Unsprung mass on front axle = 104 kg

- Using moment about the rear axle

$$GF(Adyn) = (G \times LR + m \times a \times H)/L = 1317.108N$$

- Unsprung load per wheel: $15 \times 9.81 = 147.15 \text{ N}$
- Total vertical load = 1464.25 N
- Frictional force: $0.6 \times 1464.25 = 878.258 \text{ N}$
- Braking torque:

$$T_b = \text{frictional force} \times \text{tire radius} = 878.55 \times 0.311 = 273.22 \text{ N-m}$$

Lateral Cornering Force

During cornering, centrifugal force and gyroscopic moments shift vertical load to the outer wheels. This increases the frictional force on the contact patch.

- Total vertical load (sprung + unsprung): $55 \times 9.81 = 539.55 \text{ N}$
- Gyroscopic and centrifugal load transfer:

$$F_c = [(m \times v^2 \times H) / 2 \times \text{cornering radius} \times \text{trackwidth}] = (250 \times (11.11)^2 \times 0.325) / (2 \times 3 \times 1.2) = 1392 \text{ N}$$

$$\text{Gyroscopic force: } 4I \times \omega \times \omega(p) = 1074.8 \text{ N}$$

- Net vertical load: 3026.90 N
- Lateral (frictional) force = friction coefficient $\times$ net vertical load $= 0.6 \times 3026.90 = 1816.14 \text{ N}$

$$\text{The cornering force} = 3213.43 \text{ N}$$

Bump Force

Bump forces are calculated from the vertical component of spring compression during dynamic suspension travel.

- Spring stiffness (k): 14.8 N/mm
- Max compression: 25.4 mm
- Total spring force = 381 N
- Coil spring an angle = 47°
- Vertical component:

$$F_{\text{vertical}} = 381 \times \sin(47) = 3656.77 \text{ N}$$

- Torque about suspension arm (L1 = 0.24513 m):

$$T = 3656.77 \times 0.24513 = 896.38 \text{ N}$$

- Bump force (L2 = 0.40855 m):

$$F_{\text{bump}} = \text{Torque about hinge} / \text{length of steering arm} = 896.38 \times 0.45 = 2194.05 \text{ N}$$

This bump load is applied at the suspension pickup points on the hub and contributes to deflection/stress in all three parts.

Finite Element Analysis

Finite Element Analysis (FEA) is a critical engineering tool used to simulate and predict the behaviour of components under real-world loading conditions which include stress concentration, and Total deformation.

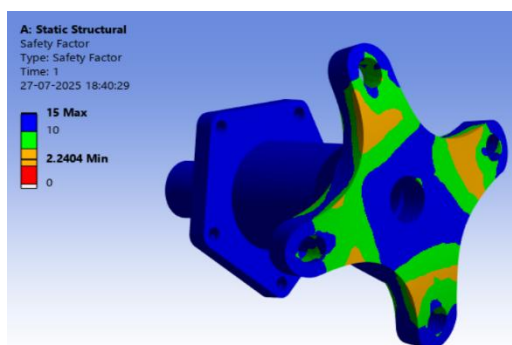


Fig 25.6 Factor of safety on rear hub

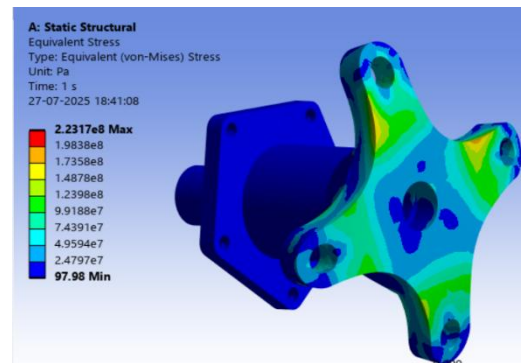


Fig 25.7 Equivalent stress on rear hub

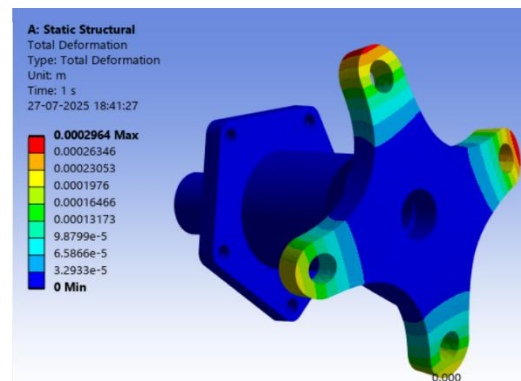


Fig 25.8 Total deformation on rear hub

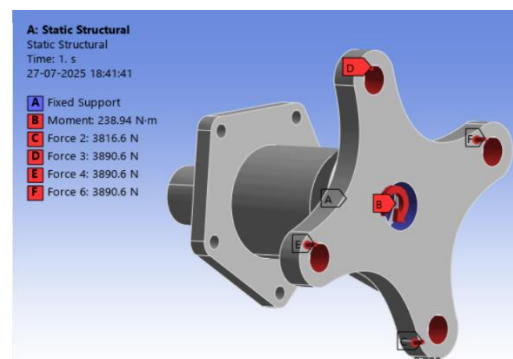


Fig 25.9 Structural analysis on rear hub

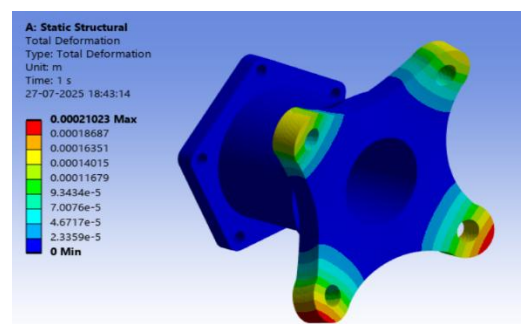


Fig 25.9 Total deformation in front hub

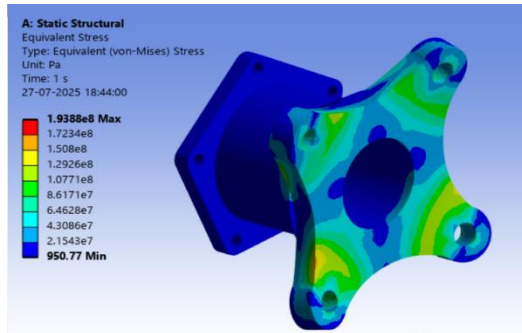


Fig 25.10 Equivalent stress on front hub

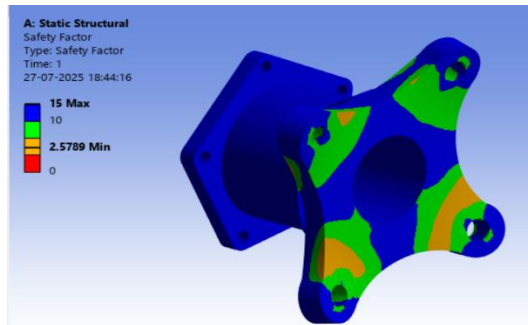


Fig 25.11 Factor of safety on front hub

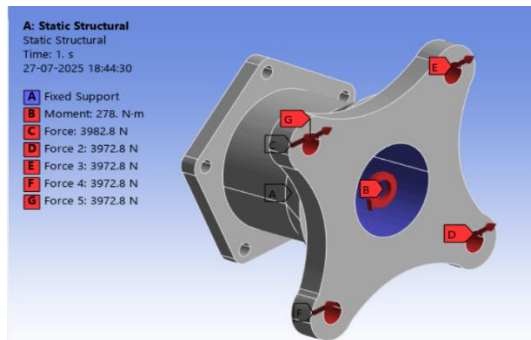


Fig 25.12 Structural analysis on front hub

Result

Rear Hub

- Maximum Equivalent (Von Mises) Stress: 223 .08 MPa
- Maximum Total Deformation: 0.263 mm
- Factor of Safety: 2.24

Front Hub

- Maximum Equivalent (Von Mises) Stress: 193.56 MPa
- Maximum Total Deformation: ~ 0.21 mm
- Factor of Safety: 2.57

Significance of the Results:

1. Equivalent Stress (Von Mises):

- Both front and rear hub stresses are well below the yield strength of Aluminium 7075-T6, ensuring elastic behaviour under all racing conditions.

2. Total Deformation:

- Reflects structural stiffness. The very low deformation values (<0.3 mm) demonstrate excellent rigidity, ensuring precise handling and minimal deflection under braking, cornering, or bump forces.

3. Factor of Safety (FOS):

- A high FOS (>1.5) confirms the design is safe and conservative, withstanding worst-case scenarios without material failure.

Conclusion

The CAE-based design and optimization of the front and rear hubs for a Formula SAE car have resulted in structurally efficient components with high safety margins.

RACK AND PINION ANALYSIS

Symbols and their values

Mass of vehicle, M	250 kg
Mass of front of vehicle, M _f	114 kg
Mass of rear of vehicle, M _r	136 kg
Wheelbase, L	1625 mm
Distance of centre of gravity from rear axle, L _r	741.46 mm
Scrub radius, r _s	63.77 mm
Kingpin inclination, λ	-3.61°
Steer angle, δ	30.08°
Caster angle, ν	2.6°
Radius of tire, r	228.6 mm
Coefficient of friction, μ	0.72
Height of centre of gravity above ground	325 mm
Length of steering arm, L _s	45 mm
Diameter of pinion, d	22.5 mm

Young's Modulus	205 GPa
Yield Tensile Strength	680 MPa
Ultimate Tensile Strength	925 MPa
Poisson's ratio	0.3

Analysis Methodology

Finite Element Analysis (FEA) was performed on the rack and pinion arrangement using Ansys software to simulate the effect of the forces and moments which are acting on it.

Meshing was done with adaptive sizing and element size was set to 1 mm. Tetrahedral elements were chosen for meshing since a tetrahedral mesh has the advantage of being able to well approximate the surface contour. The number of elements in the mesh were 23,691 and the number of nodes were 43,829.

Objective

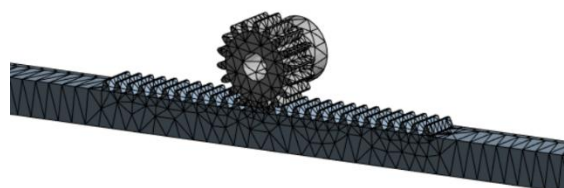
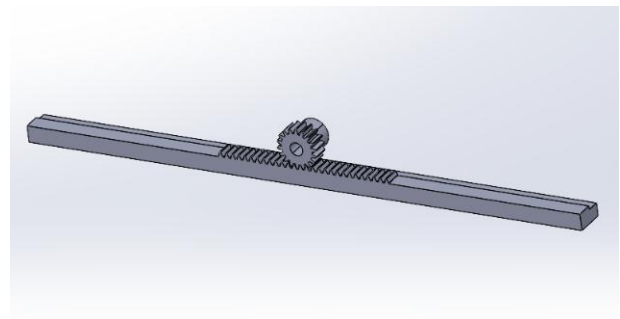
The rack and pinion is a mechanism that converts rotational motion of the steering wheel to linear motion of the rack. It is a fundamental component of the steering mechanism and connects the wheel to the rest of the vehicle through the tie-rod.

The rack and pinion must be strong enough to withstand the steering torque and the forces encountered during steering. Besides this, the rack and pinion material should also be wear resistant.

Design specification

EN24T Steel was selected for fabricating the rack and pinion due to its excellent strength and wear resistance.

The properties of EN24T Steel are:



CALCULATIONS

MOMENT DUE TO VERTICAL LOAD (Mv)

Mass of the vehicle = 250 kg

Mass of front of vehicle = 114 kg

Weight on each front wheel is $M_f \cdot g / 2 =$

559.2 N

Load transfer during cornering (F) =
 $M \cdot 1.4g \cdot h/t = 382.59 \text{ N}$

Load Transfer During Cornering on front axle =
 $F \cdot L_r/L = 174.46 \text{ N}$

Let us assume that vehicle is taking a turn
 from right to left then weight transfer takes
 place from right to left.

Rfl = Vertical reaction of front left wheel

Rfr = Vertical reaction of front right wheel.

$Rfl = 559.2 + 174.46 = 733.66 \text{ N}$

$Rfr = 559.2 - 174.46 = 384.74 \text{ N}$

$M_v = - (Rfl + Rfr) \cdot (r_s \cdot \sin \lambda \cdot \sin \delta) +$

$(Rfl - Rfr) \cdot (r_s \cdot \sin v \cdot \cos \delta)$

$M_v = 3124.035 \text{ N-mm}$

MOMENT DUE TO LATERAL LOAD (Ml)

Rfl = Lateral reaction of front left wheel

Rfr = Lateral reaction of front right wheel

$Rfl = Rfr = 1.5 \cdot M \cdot g / 4 = 919.69 \text{ N}$

$Ml = -(Rfl + Rfr) \cdot (r \cdot \tan(v))$

$Ml = 19094.34 \text{ N-mm}$

MOMENT DUE TO TRACTIVE LOAD (Mt)

Rfl = Tractive reaction of front left wheel

Rfr = Tractive reaction of front right wheel

$Rfl = \text{Vertical reaction on left wheel} \cdot \mu$

$= 733.635 \cdot 0.72 = 528.217 \text{ N}$

$Rfr = \text{Vertical reaction on right wheel} \cdot \mu$

$= 384.711 \cdot 0.72 = 276.999 \text{ N}$

$M_t = (Rfl - Rfr) \cdot r_s$

$M_t = 16020.64 \text{ N-mm}$

Total Moment on Steering Axis, $M = M_v + Ml +$
 $M_t = 38239.02 \text{ N-mm}$

In the extreme case considered, tie rod is
 inclined $\sim 60^\circ$ to rack on inner side
 of turn and $\sim 52.3^\circ$ on outer side on
 turn. These angles were obtained on
 simulating the steering conditions on LOTUS
 software.

The force on rack is sum of two forces to be
 exerted on two tie rods, force along tie-rod
 acts a moment on steering axis.

Let force on Rack $F = F_1 + F_2$

Moment on steering Axis $M = (u^r) \cdot (r \times F)$

$M = (u^r \cdot \hat{z}) \cdot (L_s \cdot F)$

$(u^r) \cdot \hat{z} = 0.97$

$M = 0.97 \cdot (45 \cdot F_1 \cdot \cos 60^\circ)$

$M = 0.97 \cdot (45 \cdot F_2 \cdot \cos 52.3^\circ)$

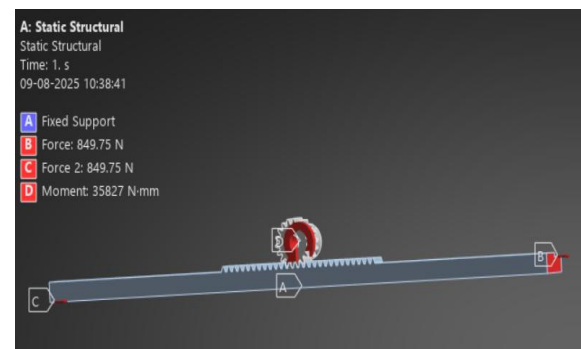
$F = F_1 + F_2 = 1752.07 + 1432.54$

$F = 3184.614 \text{ N}$

Torque on Pinion = $F \cdot (d/2)$

Torque on Pinion = 35826.9 N-mm

Force on tie rod = $M_t / L_s = 849.756 \text{ N}$



Results

The maximum equivalent von mises stress was 268.12 MPa and was obtained at the point where teeth of rack and pinion mesh with each other. The maximum deformation was 0.082 mm. The factor of safety exceeds 2.53.

This indicates that the rack and pinion design is safe and capable of handling the forces and moments encountered during steering without risk of failure.

